Distributed Service Migration in Satellite Mobile Edge Computing

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***Abstract*—With the emergence of more and more latency-sensitive applications and mobile devices pumped to the edge of network, the burden on the backhaul network is getting heavier and heavier due to the limited transmission resources. Mobile edge computing (MEC) considered as a promising technology becomes more and more popular, which can provide services at the edge of the network. In this paper, we take into account the mobility of users and focus on the problem of service migration. Most of existing works modeled a Markov Decision Process (MDP) model with a high-dimensional state space, and have to solve it by deep reinforcement learning. To tackle this issue, we propose a distributed two-layer decomposition model and generate a series of new MDP problem with Low-dimensional in order to replace original High-dimensional MDP. In our model, the size of state space is reduced from M^{2N} to $N \times M^2$ by decomposing the original optimization problem. Simulation results show that the performance of the proposed two-layer decomposition model is better than the baseline models.**

***Index Terms*—Markov Decision Process; service migration; two-layer decomposition; mobility**

I. Introduction

With the rapid popularity of mobile application, a large amount of emerging services have appeared at the edge of network, especially for virtual reality (VR), augmented reality and interactive game, which occupy quite a few CPU resources. Above mobile applications can request CPU resources from the cloud center. However, the backhaul network can't handle such a large number of computation-intensive and delay-sensitive requests with the stable transmission delay from remote clouds, and the inevitable delay jitter will make the service fail. Mobile edge computing (MEC) considered as a promising technology appears, which can offload services to edge servers and calculate the demand of users at the edge of the network. This alternative solution significantly reduces transmission delays and guarantees the QoS of dissimilar mobile applications.

Although mobile edge computing bring countless advantages including the improvement of QoS, it also creates quite a few challenges. One of the most noteworthy challenges is to consider the mobility of users, which causes a classical problem named service migration. Considering that a user requests a service from the BS that he accesses, if the user moves to another BS that is far from the original BS, then how should the service move? In this process, what we need to do is to decide whether to migrate the service instance and where to migrate in real time.

In the literature, to cope with such a challenge in satellite edge networks, there are plenty of references dedicated to solving this problem. [1] was proposed as the first paper concerning service migration in 2013, and a Markov chains model based on the distance was put forward to solve the mobility of users. Then, [2] raised Markov Decision Process model to depict service migration with decision and action, while the distance between user and service was defined as the state of this MDP model. However, users moved on a one-dimensional plane in the literature mentioned above. Thus, [3] improved the aforementioned shortcomings and proposd a two-dimensional mobility, which made the MDP model more actual. Above papers only considered single service and single user causing a small state space without the problem of curse of dimensionality, which lacked the constraints on remaining resources at the base station comparing with multi-service and multi-user model. Thus, [4] proposed a multi-service and multi-user model with a four-tuple defined as system state, which can constraint BS resources effectively. A Lyapunov optimization was proposed in this paper to solve the problem of high-dimensional state space. While another alternative solution was DQN proposed in [5], which can also solve the problem of dimensionality curse in multi-service and multi-user model, and the DQN was more universal compared with other methods. In addition, there are also some novel paper, for example [6] proposed POMDP model which is different from traditional MDP and [7] considered elastic services, that is, different service instances can be applied to the same user with different QoS.

Given the above, the literature of service migration based on MDP can be divided into two categories, one is single service model, this model lacks CPU resources constraints for each BS when multi-service runs at the same time. The other is multi-service model, this model has the problem of dimensionality curse, which can be solved by DQN or Lyapunov optimization. Motivated by above paper, we proposed a two-layer decomposition method that can relieve the curse of dimensionality about the multi-service model. The core idea of the decomposition in this article is the mutual conversion between optimization problems and MDP problems. Thus, the original MDP problem

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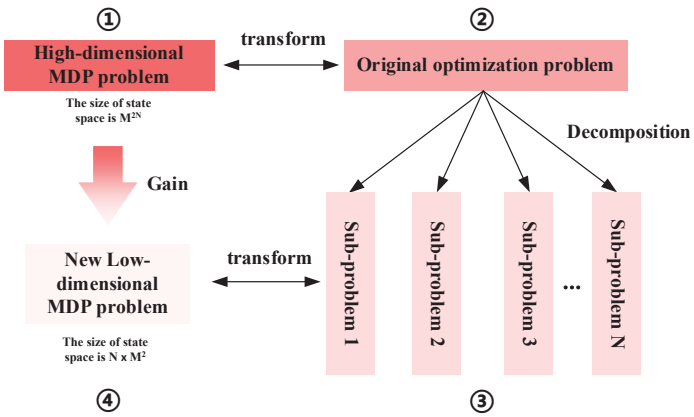

Fig. 1: The illustration of the method proposed in our paper that can relieve the curse of dimensionality by decomposition.

which has a huge state space can be depicted as an optimization problem. Then, we decompose this optimization problem by the two-layer decomposition, and obtain N subproblems. Finally, we turn those subproblems into new MDP problems, where each new MDP has a low dimensionality state space. This process can be illustrated as Fig. 1. The contributions of this paper are listed as follows:

- An online optimization model considered the mobility of user is proposed in our paper based on MDP, which can solve the problem of service migration. Our service migration model focuses on multi-service with the scenario of limited BS resources.
- A distributed two-layer decomposition method was proposed, which can realize parallel optimization and reduce computation complexity compared with centralized optimization. As far as we know, our two-layer decomposition model was first applied to the service migration problem.

The rest of this paper is organized as follows. The system model and symbol description is introduced in section II. A two-layer decomposition method is proposed in section III. While in section IV, the low-dimensional MDP is established and the decomposition algorithm is depicted. Finally, the simulation results and the conclusion are given in section V and section VI, respectively.

II. System Model

In this section, we will introduce our system model and symbol description. We assume random mobile users offload latency-sensitive tasks to the edge servers (BSs) endowed with limited CPU resources. Let $\mathcal{M} = \{1, 2, \cdots, M\}$ denote the set of BSs, and $\mathcal{N} = \{1, 2, \cdots, N\}$ denote the set of users. Considering that each user needs a service to satisfy his requirement, thus the number of user equals to the number of service. A three-tuple $V_n = (I_n^t, \phi_n^t, D_n^t)$ is proposed to depict different services, where I_n^t denotes the data size of service instance n that need to migrate, ϕ_n^t represents how many CPU cycles are requested in time slot t for service n, meaning that the total CPU cycles $\Phi_n = \sum_{t=1}^{T} \phi_n^t$, D_n^t denotes the size of remote communication in time slot t if the service and the user are not located in the same BS. The set of decision variables denote as (X, C, W, P), where $X = \{x_n^t, x_n^t \in \mathcal{M}\}$ denotes the decision of migration. For example, $x_n^t = m$ means service n migrates to BS m at the beginning of time slot t. Similarly, we use $U = \{u_n^t, u_n^t \in \mathcal{M}\}$ denotes user's location, $u_n^t = k$ means user n randomly moves to BS k in the time slot t. Then, $C = \{C_n^t, C_n^t \in [C_n^{low}, C_n^{up}]\}$, $W = \{W_n^t, W_n^t \in [W_n^{low}, W_n^{up}]\}$, $P = \{P_n^t, P_n^t \in [P_n^{low}, P_n^{up}]\}$ denote the set of CPU resources, bandwidth resources and power resources which occupied by the service n in time slot t, respectively.

Next, we will introduce three types of delay as the objective function of our optimization problem. The first is calculation delay that is given as follow.

$$T_n^s = \sum_{t=1}^{T} T_n^s(t) = \sum_{t=1}^{T} \frac{\phi_n^t}{C_n^t}, \tag{1}$$

The second delay is communication delay consisting of two parts, that is, the access delay $T_n^{access}(t)$ and the backhaul delay $T_n^{backhaul}(t)$.

$$T_n^c = \sum_{t=1}^{T} T_n^c(t) = \sum_{t=1}^{T} (T_n^{access}(t) + T_n^{backhaul}(t)), \tag{2}$$

where $T_n^{access}(t)$ and $T_n^{backhaul}(t)$ are given as follow.

$$T_n^{access}(t) = \frac{D_n^t}{W_n^t log_2(1 + \frac{P_n^t H_{mn}^t}{\sigma_m^2})}, \tag{3}$$

where H_{mn}^t denotes channel gain, and σ_m^2 denotes white noise power. The denominator of this formula represents communication rate [5], thus $T_n^{access}(t)$ denotes the access delay between the user and the BS in time slot t.

$$T_n^{backhaul}(t) = \begin{cases} 2mh_{x_n^t, u_n^t} f, & if\ x_n^t \neq u_n^t \\ 0, & otherwise \end{cases} \tag{4}$$

where $h_{x_n^t, u_n^t}$ denotes the number of hop between BS x_n^t and BS u_n^t, m is a weight coefficient, f denotes the times of communication between users and service instances in a time slot.

The third delay is migration delay, which only occurs if the service migrates to other BSs, meaning that

$$T_n^m = \sum_{t=1}^{T} (I_{\{x_n^{t-1} \neq x_n^t\}} (\frac{I_n^t}{\rho_{x_n^{t-1}, x_n^t}} + t_0)), \tag{5}$$

where $\rho_{x_n^{t-1}, x_n^t}$ is the transmission rate between BS x_n^{t-1} and BS x_n^t, if x_n^{t-1} equals to x_n^t, then $T_n^m(t) = 0$. In addition, the constant t_0 means the migration time cost consisted of two parts, one is the downtime for service n in BS x_n^{t-1}, the other is the uptime for service n in BS x_n^t.

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The objective function is shown as $T_{total} = \sum_{n=1}^{N} \xi_n (T_n^s + T_n^c + T_n^m)$. As such, our optimization problem can be formulated as follow.

$$\boldsymbol{P_1}: \min_{X,C,W,P} T_{total} = \sum_{n=1}^{N} \xi_n (T_n^s + T_n^c + T_n^m), \tag{6}$$

$$s.t. \quad \begin{cases} X \in \mathcal{M} \\ \sum_{n=1}^{N} I_{\{x_n^t = m\}} C_n^t \le C_0, \quad \forall t, \forall m \\ \sum_{n=1}^{N} I_{\{x_n^t = m\}} W_n^t \le W_0, \quad \forall t, \forall m \\ \sum_{n=1}^{N} I_{\{x_n^t = m\}} P_n^t \le P_0, \quad \forall t, \forall m \\ C_n^{low} \le C_n^t \le C_n^{up}, \quad \forall t \\ W_n^{low} \le W_n^t \le W_n^{up}, \quad \forall t \\ P_n^{low} \le P_n^t \le P_n^{up}, \quad \forall t \end{cases} \tag{7}$$

where $I_{\{x_n^t = m\}}$ denotes indicative function, demonstrating that $I_{\{x_n^t = m\}} = 1$ if $x_n^t = m$ and $I_{\{x_n^t = m\}} = 0$ otherwise. While ξ_n denotes the importance of service n or weight coefficient of service n. Let's set $S^a = \{\xi_n, C_n^{low}, C_n^{up}, W_n^{low}, W_n^{up}, P_n^{low}, P_n^{up}, \forall n\}$ denote the parameters that controlled by satellites. Parameter S^a determines the weight of a service as well as the resources which a service can obtain meaning that each service can be controlled by satellites. The difficulty of this problem is that each user's movement is random, that is, u_n^t follows a random walk model on a two-dimensional plane.

Considering that the above problem is mixed-integer programming, which is difficult to solve, especially for the integer variable X. The traditional solution for X is to directly build the MDP model with a high-dimensional state space. For example, $S_0^t = (x_1^t, x_2^t, x_3^t, \ldots, x_N^t; u_1^t, u_2^t, u_3^t, \ldots, u_N^t)$ is the system state at t time slot. Obviously, the size of S is M^{2N}, which is an extremely huge number and cannot be solved directly by traditional value iteration, policy iteration or Q-learning due to the curse of dimensionality. As such, a two-layer decomposition model is proposed in section III to solve the above problem.

III. Two-layer decomposition

We decouple problem P_1 in two steps, that is, direct decomposition and dual decomposition. Direct decomposition focuses on decoupling BS and variables, which achieves the transformation from centralized optimization to distributed BS optimization.

A. Direct decomposition

For the sake of simplicity, we assume $\xi_n = 1$ for any $n \in \mathcal{N}$. The objective function can be rewrite as follow,

$$T_{total} = \sum_{m=1}^{M} (\sum_{n=1}^{N} I_{\{x_n^t = m\}} (T_n^s + T_n^c + T_n^m)) \tag{8}$$

Next, problem P_1 can be transformed to problem P_2 which aims at optimizing variables C, W, P and problem P_3 which aims at optimizing variables X by direct decomposition.

$$\boldsymbol{P_2}: \min_{C,W,P} T_{total}(X(C,W,P), C, W, P) \tag{9}$$

$$s.t. \quad \begin{cases} \sum_{n=1}^{N} I_{\{x_n^t = m\}} C_n^t \le C_0, \quad \forall t, \forall m \\ \sum_{n=1}^{N} I_{\{x_n^t = m\}} W_n^t \le W_0, \quad \forall t, \forall m \\ \sum_{n=1}^{N} I_{\{x_n^t = m\}} P_n^t \le P_0, \quad \forall t, \forall m \\ C_n^{low} \le C_n^t \le C_n^{up}, \quad \forall t \\ W_n^{low} \le W_n^t \le W_n^{up}, \quad \forall t \\ P_n^{low} \le P_n^t \le P_n^{up}, \quad \forall t \end{cases} \tag{10}$$

It is evident that the above problem is a convex optimization problem, where C, W, P are continuous variables. The other part of the original problem is to optimize the variable X, which is an integer programming problem. We directly decouple the optimization problem for each BS m and get the following subproblem P_3.

$$\boldsymbol{P_3}: \min_{X} \sum_{n=1}^{N} I_{\{x_n^t = m\}} (T_n^s + T_n^c + T_n^m), \quad \forall m \tag{11}$$

$$s.t. \quad \begin{cases} X \in \mathcal{M} \\ \sum_{n=1}^{N} I_{\{x_n^t = m\}} C_n^t \le C_0, \quad \forall t \\ \sum_{n=1}^{N} I_{\{x_n^t = m\}} W_n^t \le W_0, \quad \forall t \\ \sum_{n=1}^{N} I_{\{x_n^t = m\}} P_n^t \le P_0, \quad \forall t \end{cases} \tag{12}$$

B. Dual decomposition

So far, we have completed the first-layer of decomposition, then we will introduce the second-layer decomposition, that is, dual decomposition [8]. First, let's get the Lagrangian function of problem P_3 as follow.

$$\begin{aligned} L(X, \lambda, \mu, \eta) &= \sum_{n=1}^{N} I_{\{x_n^t = m\}} (T_n^s + T_n^c + T_n^m) + \\ &\sum_{t=1}^{T} \lambda_t (\sum_{n=1}^{N} I_{\{x_n^t = m\}} C_n^t - C_0) + \sum_{t=1}^{T} \mu_t (\sum_{n=1}^{N} I_{\{x_n^t = m\}} P_n^t - \\ &P_0) + \sum_{t=1}^{T} \eta_t (\sum_{n=1}^{N} I_{\{x_n^t = m\}} W_n^t - W_0) \\ &= \sum_{n=1}^{N} I_{\{x_n^t = m\}} (T_n^s + T_n^c + T_n^m + \sum_{t=1}^{T} \lambda_t C_n^t + \sum_{t=1}^{T} \mu_t P_n^t \\ &+ \sum_{t=1}^{T} \eta_t W_n^t) - (\sum_{t=1}^{T} \lambda_t C_0 + \sum_{t=1}^{T} \mu_t P_0 + \sum_{t=1}^{T} \eta_t W_0) \\ &= \sum_{n=1}^{N} I_{\{x_n^t = m\}} g_n(X, \lambda, \mu, \eta) - K(\lambda, \mu, \eta) \end{aligned} \tag{13}$$

where $g_n(X, \lambda, \mu, \eta) = T_n^s + T_n^c + T_n^m + \sum_{t=1}^{T} (\lambda_t C_n^t + \mu_t P_n^t + \eta_t W_n^t)$ is a new objective function including Lagrange multiplier

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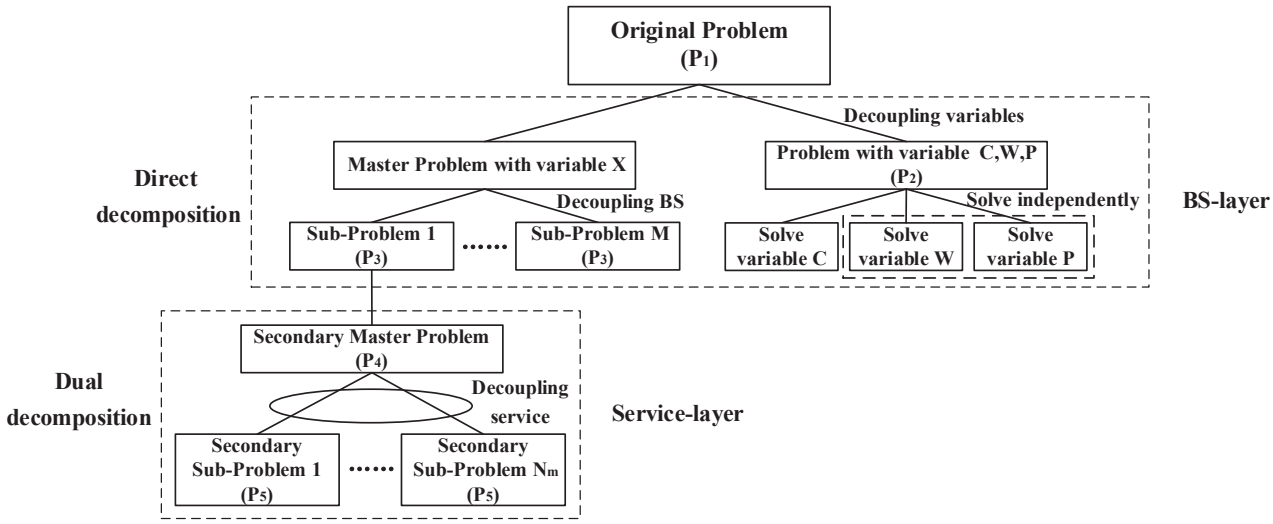

Fig. 2: Detailed decomposition architecture of the two-layer decomposition. The number of sub-problem P_3 is M, and the number of secondary sub-problem P_5 is N, where $N = \sum_{m=1}^{M} N_m$ and N_m is the number of service located in BS m.

λ, μ, η, and $K(\lambda, \mu, \eta) = \mathbf{1}\lambda^T C_0 + \mathbf{1}\mu^T P_0 + \mathbf{1}\eta^T W_0$ is the part without X. Following the preceding basic rationale, we can rewrite problem P_3 by dual decomposition as a master problem P_4 and multi-subproblems P_5 as follow.

$$\boldsymbol{P_4}: \max_{\lambda,\mu,\eta} \sum_{n=1}^{N} I_{\{x_n^t = m\}} g_n(\lambda, \mu, \eta) - K(\lambda, \mu, \eta) \tag{14}$$

$$s.t. \lambda \succeq 0, \mu \succeq 0, \eta \succeq 0 \tag{15}$$

where problem P_4 is the master problem, which aims at optimizing Lagrange multiplier λ, μ, η.

$$\boldsymbol{P_5}: \min_{X} T_n^s + T_n^c + T_n^m + \sum_{t=1}^{T} (\lambda_t C_n^t + \mu_t P_n^t + \eta_t W_n^t), \tag{16}$$

$$s.t. \quad X \in \mathcal{M} \tag{17}$$

where problem P_5 is the sub-problem for each service. The second decomposition achieves the transformation from distributed BS optimization to distributed service optimization, demonstrating that problem P_5 aims at optimizing each service and problem P_3 aims at optimizing each BS. It is evident that the granularity between problem P_5 and problem P_3 is different.

C. Decomposition architecture

As a final remark, we will give the decoupled architecture consisting of BS-layer and service-layer about our problem in Fig. 2, where the BS-layer corresponds to direct decomposition and the service-layer corresponds to dual decomposition. In the BS-layer, the smallest optimization unit is BS, as we can see that the objective function in formula 11 is the sum of delay for the services which are located in each BS. While in the service-layer, the smallest optimization unit is service, as we can see that the objective function in formula 16 is the delay for only one service with a Lagrangian remainder.

IV. New MDP Problem

In this subsection, we will introduce our new MDP model with the advantage of low-dimensionality. Firstly, let's review the original MDP that is induced by original problem P_1. If we define the MDP model according to the problem P_1, then the system state contains the location of all users and the location of all services, which means $S_0^t = (x_1^t, x_2^t, x_3^t, \ldots, x_N^t; u_1^t, u_2^t, u_3^t, \ldots, u_N^t)$, where the size of state space S_0 is M^{2N}. If $M = 10$, $N = 20$, then $M^{2N} = 10^{40}$ which is difficult to calculate. Now, let's define the state of new MDP model for service n at time slot t as $S_n^t = \{x_n^t, u_n^t\}$ with the size of M^2 according to problem P_5 instead of problem P_1. While the action is defined as the decision of migration, that is, $a_n^t = x_n^t$. It is worth mentioning that a_n^t occurs at the beginning of time slot t, then the state of service n turns from S^{t-1} to $\hat{S}^{t-1}$. After the user is transferred, system status changes from $\hat{S}^{t-1}$ to S^t. Next, we will introduce the model of users' mobility. The mobility of users is defined as a random walk model among BSs, each user has an equal probability to transfer to its neighbor BS or stay in the local BS. Let $\mathcal{F}_n^t = \{\{u_n^{t-1}\} \cup \mathcal{N}_n^{t-1}\}$ denotes the feasible migration BS at the time slot t for user n, which includes BS u_n^{t-1} and its neighbor BS $\mathcal{N}_n^{t-1}$. For the sake of simplicity, The transition probability of user n can be depicted as follows, $p(u_n^t = f_{bs}) = \frac{1}{|\mathcal{F}_n^t|}$, where $f_{bs} \in \mathcal{F}_n^t$.

Now, let's derive the objective function of P_5 for each slot in order to define the reward matrix.

$$g_n(X, \lambda, \mu, \eta) = T_n^s + T_n^c + T_n^m + \sum_{t=1}^{T} (\lambda_t C_n^t + \mu_t P_n^t + \eta_t W_n^t)$$

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Algorithm 1 Two-layer decomposition algorithm

Input: S^a, V_n.
Output: X, C, W, P.

1: **Initialize** $X, C, W, P, \lambda, \mu, \eta$.
2: **for** $t = 1, 2, ..., T$ **do**
3: **while** (1) **do**
4: **for** m=1,2,...,M **do**
5: Solve convex optimization problem P_2 with respect to the variable C.
6: **while** (1) **do**
7: Solve convex optimization problem P_2 with respect to the variable P.
8: Solve convex optimization problem P_2 with respect to the variable W.
9: **if** $||P(n_1)-P(n_1-1)|| < \epsilon_1 \& ||W(n_1)-W(n_1-1)|| < \epsilon_1$ **then**
10: break
11: **end if**
12: **end while**
13: **while** (1) **do**
14: **for** $k = 1, 2, ..., N_m$ **do**
15: Train the MDP model with decision variables and Lagrange multiplier by policy iteration.
16: Solve P_5 with respect to the variable X for each service k that located in BS m by MDP.
17: **end for**
18: //Update Lagrange multiplier
19: $\lambda(n_2) = \lambda(n_2 - 1) + \alpha(\sum_{n=1}^{N_m} C_n^t - C_0)^+$
20: $\mu(n_2) = \mu(n_2 - 1) + \alpha(\sum_{n=1}^{N_m} P_n^t - P_0)^+$
21: $\eta(n_2) = \eta(n_2 - 1) + \alpha(\sum_{n=1}^{N_m} W_n^t - W_0)^+$
22: **if** $|| \sum_{n=1}^{N_m} g_n^t(n_2) - \sum_{n=1}^{N_m} g_n^t(n_2 - 1)|| < \epsilon_2$ **then**
23: break //Convergence of the objective function
24: **end if**
25: **end while**
26: **end for**
27: **if** $||P(n_3) - P(n_3 - 1)|| < \epsilon_1 \& ||W(n_3) - W(n_3 - 1)|| < \epsilon_1 \& ||C(n_3) - C(n_3 - 1)|| < \epsilon_1 \& (X(n_3) == X(n_3 - 1))$ **then**
28: break
29: **end if**
30: **end while**
31: **end for**
32: **return** X, C, W, P.

$$= \sum_{t=1}^{T} (T_n^s(t) + T_n^c(t) + T_n^m(t) + \lambda_t C_n^t + \mu_t P_n^t + \eta_t W_n^t)$$
$$= \sum_{t=1}^{T} g_n^t \tag{18}$$

where $g_n^t = T_n^s(t) + T_n^c(t) + T_n^m(t) + \lambda_t C_n^t + \mu_t P_n^t + \eta_t W_n^t$ denoted the cost of service n at time slot t including total delay $T_n^s(t) + T_n^c(t) + T_n^m(t)$ and Lagrangian factor penalty $\lambda_t C_n^t + \mu_t P_n^t + \eta_t W_n^t$. Following the preceding definition, the reward function is given as follow.

$$R_n^t(S_n^{t-1}, S_n^t) = g_n^t \Big|_{\{(S_n^{t-1}, S_n^t) = (x_n^{t-1}, u_n^t)\}} - g_n^t \Big|_{\{(S_n^{t-1}, S_n^t) = (x_n^t, u_n^t)\}} \tag{19}$$

where $R_n^t(S_n^{t-1}, S_n^t)$ represents the delay benefits of migrating agents compared to non-migrating agents.

As a final remark in this section, we will give our decomposition algorithm in Alg. 1. The input of the algorithm contains all of the parameters as elaborated above, while the output is the decision variables including X, C, W, P. The outermost loop of the algorithm is related to time slot, demonstrating that the algorithm runs at the beginning of each time slot. There are three iteration count variables n_1, n_2, n_3, which are applied to different iteration. n_1 is the count variable iterate from Line 6 to Line 12, which aims at finding the optimal P and W. n_2 is the count variable iterate from Line 13 to Line 25, which aims at optimizing X. While n_3 is the count variable iterate from Line 3 to Line 30, which aims at ensuring the convergence of decision variables X, C, W, P.

V. Simulation Results

In this section, we will give the simulation results of our model proposed above and compare the performance of our two-layer decomposition with some baseline models. First, let's give the simulation settings as the Table I listed.

TABLE I: Simulation Settings

Parameter	Value
The number of time slot T	20
The number of BS M	16
The number of service N	40
The data size of service instance I_n	[75,85] Mbps
The data size of the remote communication D_n	[1.5,2] Mbps
The number of CPU cycles requested by service n	[0.5, 0.6] Mcycles
The BS total CPU cycles C_0	3 Mcycles
The BS total bandwidth W_0	3 Mbps
The BS total power P_0	10 W
White noise power	10^{-9} W
Channel gain H_{mn}	[1.6, 1.8]$\times 10^{-7}$

It is worth mentioning that the low-dimensional MDP model bases on time slot, and the length of each time slot is 10s in our system, which is determined by user's moving speed and the coverage of the 5G BS. Concretely, the radius of the 5G base station is approximately 250 meters, if we assume that the user's moving speed is no more than 10m/s, then the average time for

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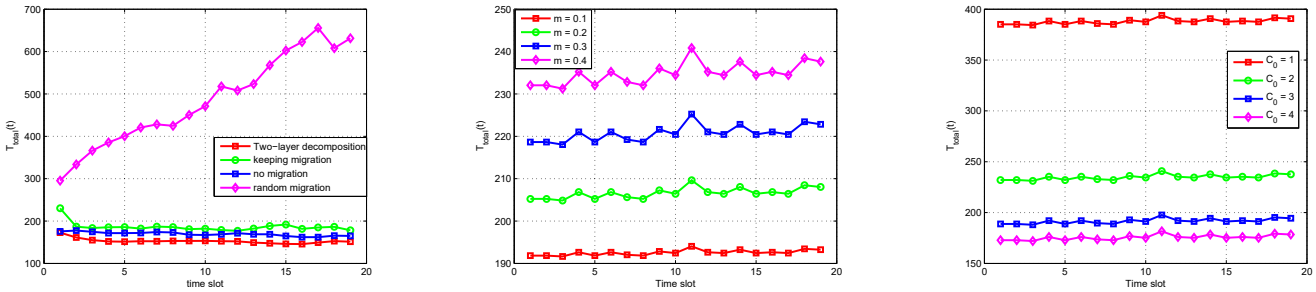

(a) The delay comparison between the two-layer decomposition model and baseline models.

(b) The influence of the coefficient m on $T_{total}(t)$

(c) The influence of the C_0 on $T_{total}(t)$

Fig. 3: Simulation results.

the user to move out of the base station is 25s if the user moves in a straight line. As such, 10s is small enough for our system. In addition, we set $C_n^{low} = 0, C_n^{up} = 2Mcycles, B_n^{low} = 0, B_n^{up} = 2Mbps, P_n^{low} = 0, P_n^{up} = 4W, \epsilon_1 = 0.001, \epsilon_2 = 0.01$.

We compare the the performance of our two-layer decomposition with three fundamental baseline models. The first baseline is random migration with respect to variable X, that is, x_n^t is random. The second baseline is keeping migration with respect to variable X according to the user's position at the time slot of $t-1$, that is, $x_n^t = u_n^{t-1}$. While the third baseline is no migration with respect to variable X, that is, x_n^t has never been changed. It is worth mentioning that random migration, keeping migration and no migration are related to variable X, while variables C, W, P in those three baseline are solved by convex optimization iteration which is same as the iterative process in algorithm 1, especially for Line 5 to Line 12.

Fig. 3 (a) illustrates the performance of our two-layer decomposition compared with random migration, keeping migration, and no migration. Considering that users can random move, and the position of users couldn't be know in advance. Therefore, in order to improve the stability of the results, we take the average of delay. This simulation is a average result considering the mobility of users, as we can see that the performance of our two-layer decomposition is better than above baselines for each time slot.

Fig. 3 (b) illustrates the influence of the coefficient m on T_{total}. It is worth mentioning that MDP model is a optimization problem, which aims at optimizing current time slot. Therefore, there are independence to some extent between time slots. In addition, we choose the value of m as 0.1, 0.2, 0.3, 0.4, respectively. It is evident that delay increases with the coefficient m, because $T_{backhaul}$ is proportional to coefficient m. Fig. 3(c) illustrates the influence of the coefficient m on T_{total}. We choose the value of C_0 as 1, 2, 3, 4, respectively. As we can seen that $T_{total}(t)$ decrease obviously when C_0 changes from 1 to 2.

VI. Conclusion

In this paper, an online optimization model based on MDP is proposed, which can solve the problem of service migration. Our service migration model focuses on multi-service with the scenario of limited BS resources. Different from traditional service migration models, we proposed a distributed two-layer decomposition model including direct decomposition and dual decomposition, which can realize parallel optimization and reduce computation complexity comparing with centralized optimization. In our model, the size of state space is reduced from M^{2N} to $N \times M^2$ by decomposing the original optimization problem. As far as we know, our two-layer decomposition model is first applied to the service migration problem.

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